



Introduction:

Face recognition systems that look near-perfect on benchmarks fail under real-world surveillance conditions: low resolution, blur, poor illumination, where a single missed identity is unacceptable. RA-FR replaces fixed Top-k retrieval with adaptive sets that provably include the ground-truth identity at a user-specified risk (α) and confidence ($1-\delta$)

Datasets:

Training: Indian Movie Facial Dataset **IMFDB**

Testing: Surveillance Camera Facial Dataset **SCFace**

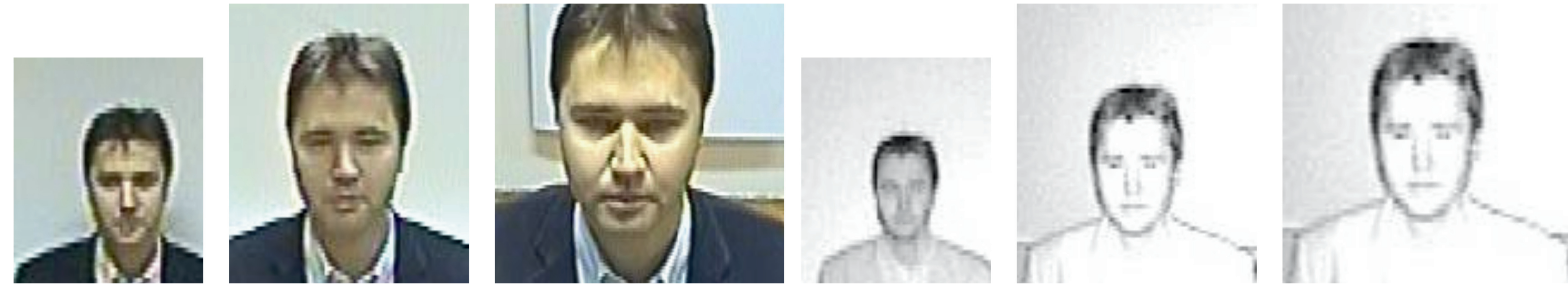


Fig1: SCFace Dataset

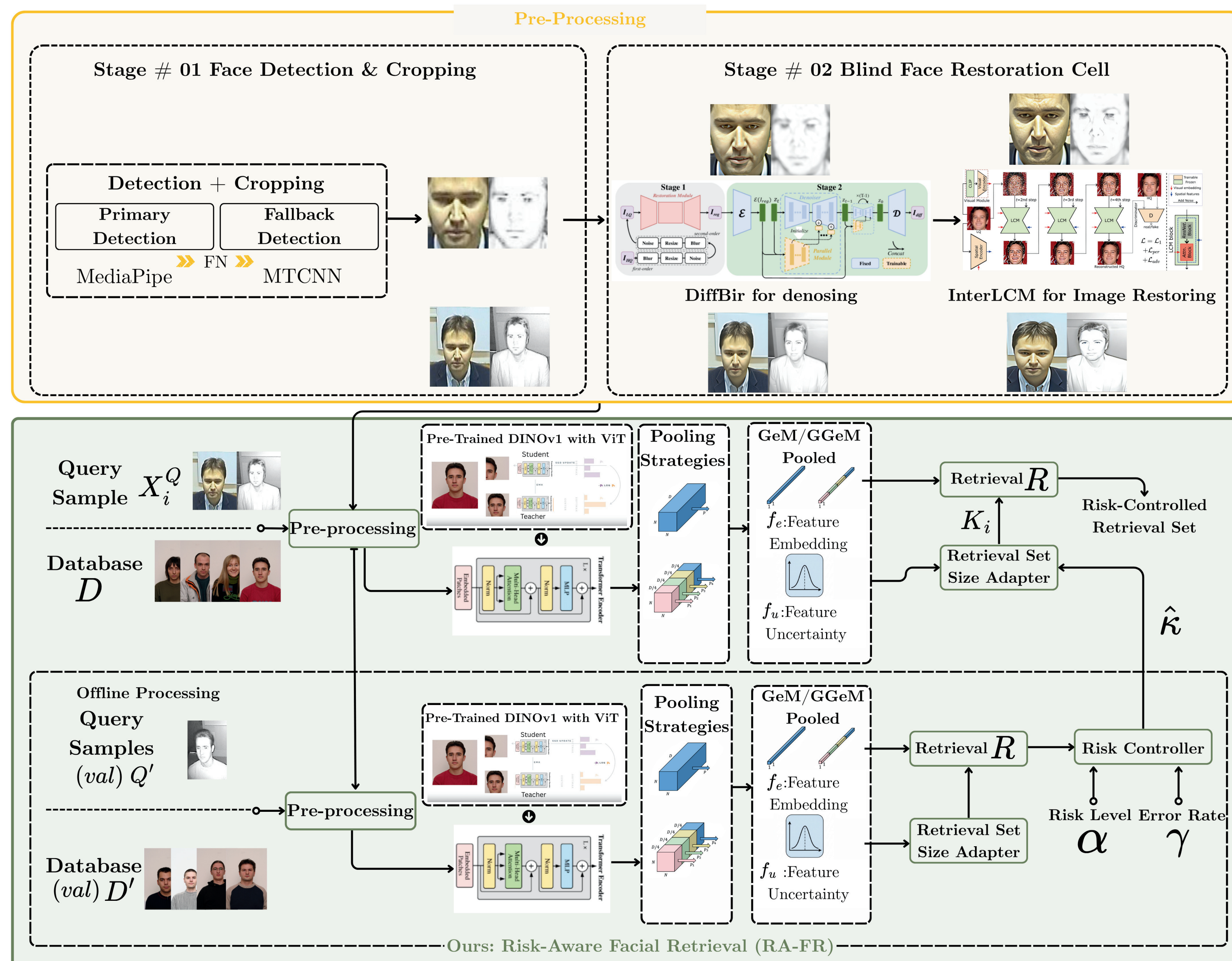


Fig. 2 RA-FR Methodology Workflow

Figure 3 illustrates the results of the RA-FR on the SCFace. The dashed line indicates the boundary between the actual risk and the required risk at α . A smaller δ value leads to more stringent results with a larger retrieval

- The red region signifies the uncontrolled area, while the green region represents the controlled area.
- RA-FR effectively manages risk across all risk values, regardless of the confidence values (δ).

Methodology:

RA-FR uses Hoeffding-bounded conformal risk control where the risk of an image retrieval system R is defined by:

$$\rho(R) = EX \in Q[\ell(R(X), S(X))]$$

and ℓ indicates the retrieval set misses all true nearest neighbors

Risk Controller: Computes offline scaling factor $\hat{k}$ using risk (α) and error (δ) to guarantee $\geq (1 - \alpha)$ coverage of true nearest neighbors.

Retrieval Set Size Adapter: Uses uncertainty $f_u(x)$ and $\hat{k}$ at inference to dynamically scale set sizes (larger for hard queries, smaller for easy ones).

Results & Discussion:

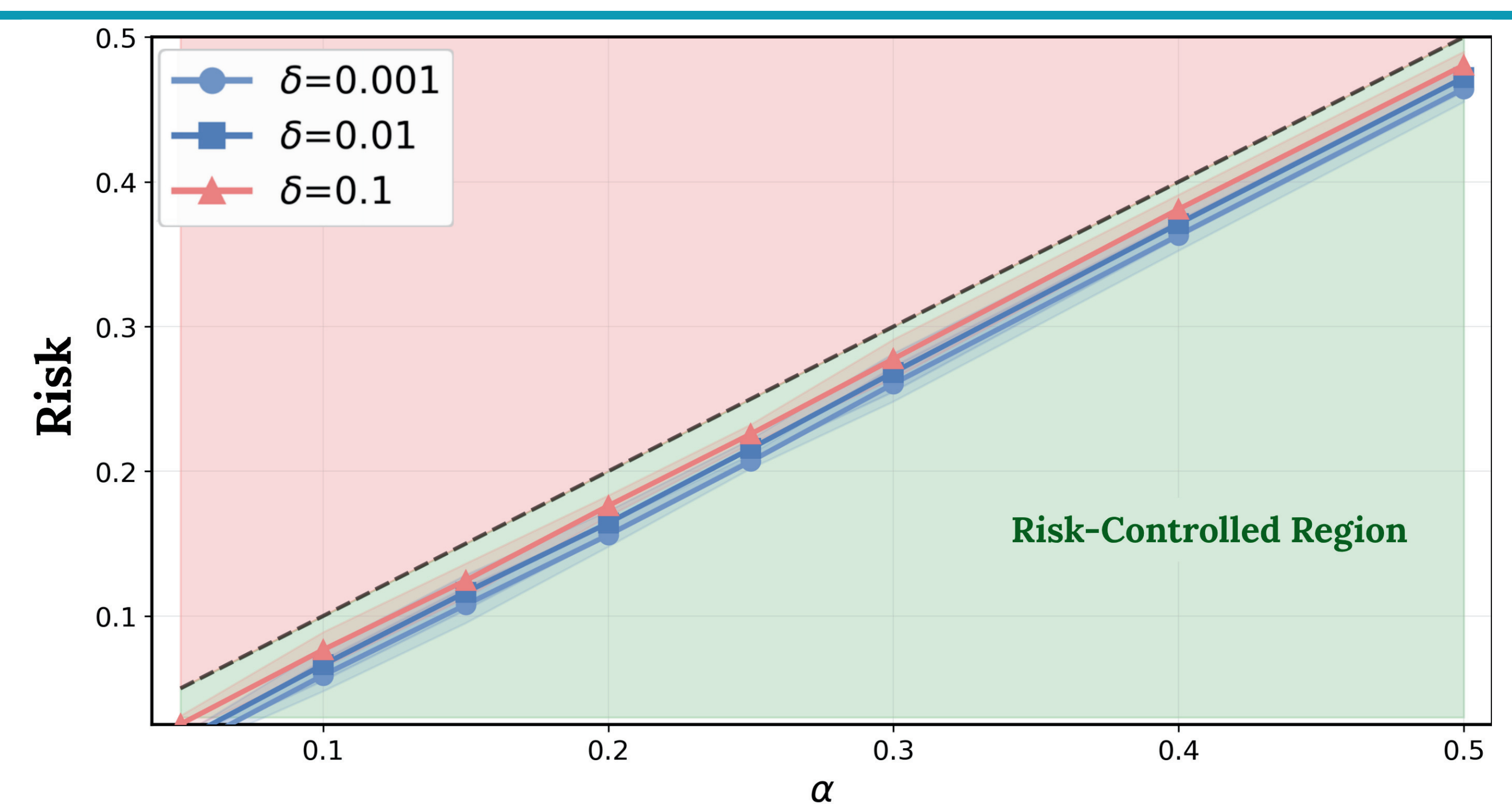


Fig.3 Empirical vs. target risk (α) on SCFace

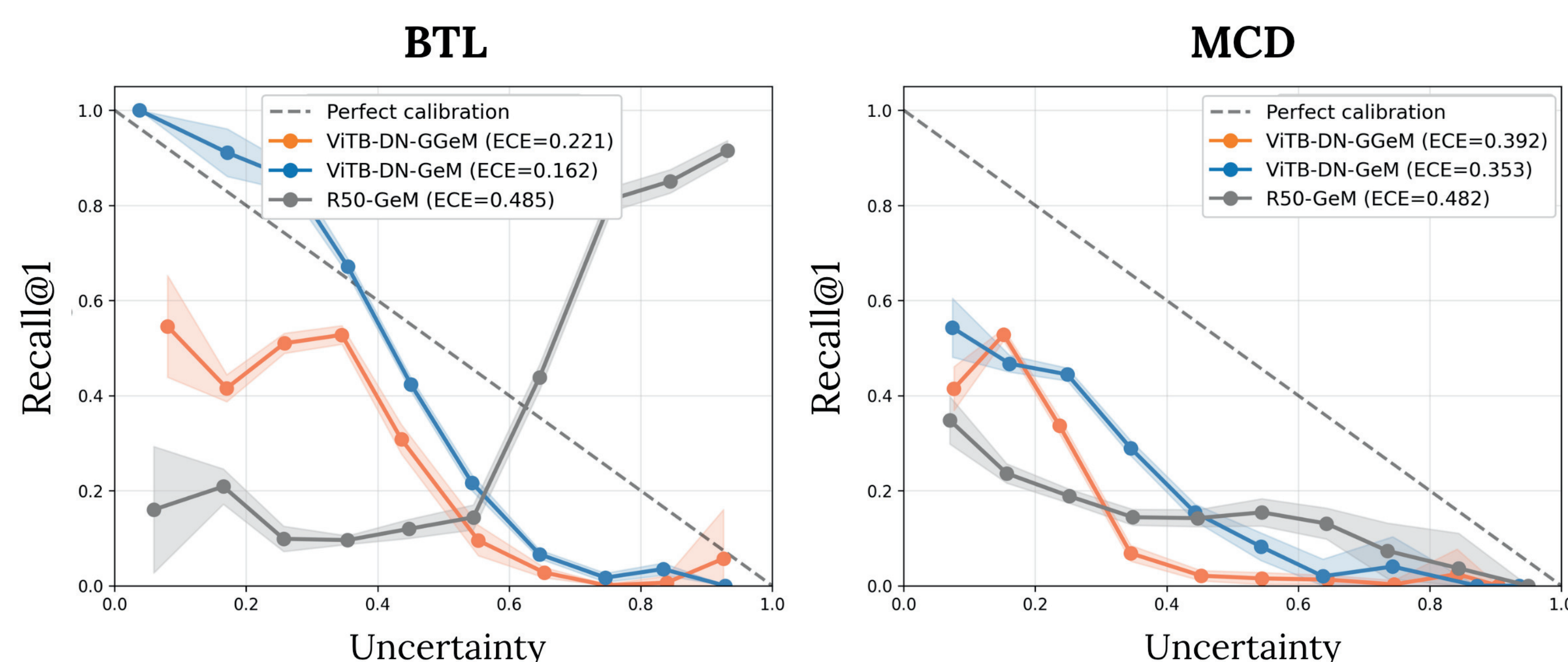


Fig.4 Expected Calibration Error (ECE) on SCFace;
(Dashed line: ideal; Shaded: std. dev.)

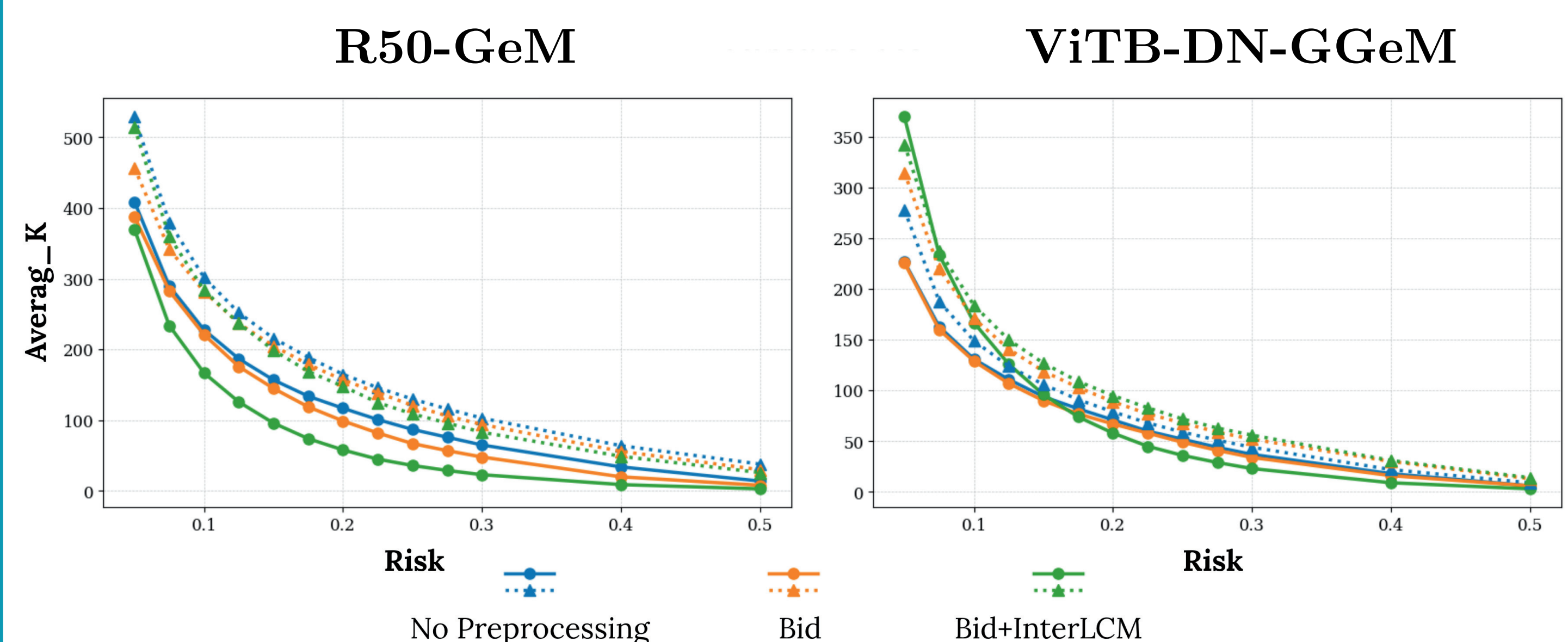


Fig.5 Avg. K vs Risk on SCFace w & w/o pre-processing, R50-GeM and proposed ViTB-DN-GGeM.

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